Self-assessment Worksheet (Worksheet 5)

Do I understand this weeks material?

Real Analysis Topics	Tick the box when you feel confident with each topic
• Understand the definition of a function and the algebraic operation on functions.	☐
• Know and understand the definition of a continuous function at a point and be able to use this to prove the continuity of specific functions.	☐
• Know and understand the definition of a discontinuous function and use it to construct proofs about discontinuous functions.	☐
• Know and understand the definition of limits, including one-sided limit, of functions and how to prove them.	☐
• Relate the definition of a limit of a function to continuity.	☐
• Know and understand the theorem on sequential continuity and use it to construct proofs.	☐
• Know and understand the intermediate and extreme value theorem.	☐

Exercises and Questions

Unless stated otherwise, you may assume any results proven in lecture.

Exercises and questions are organised and classified into three groups:

- **Starter Questions** - these questions will help cement the basic techniques.
- **Challenge Questions** - these questions are more challenging, designed to test your deeper understanding of the topic.
- **Extension Questions** - questions to try yourself in your own time to extend your knowledge and understanding.

Starter Qs

Have a go at these "warm-up" questions to get you started.

Question 1 I can do this question ☐

- Prove that the following functions are continuous at the given point directly from the $\epsilon - \delta$-definition of continuity.
  (a) $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 4x - 1$ at $x = 2$.
  (b) $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ at $x = 6$.
  (c) $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2 - 3x$ at $x = 2$.
  (d) $f : \mathbb{R} \setminus \{-5\} \to \mathbb{R}$ given by $f(x) = \dfrac{x+2}{x+5}$ at $x = 1$.

Question 2 I can do this question ☐

- For each of the following, give an example or prove no such example exists:
  (a) If $\lim_{x\to c}(f(x))$ and $\lim_{x\to c}(g(x))$ do not exist, can $\lim_{x\to c}(f(x) + g(x))$ exist?
  (b) If $\lim_{x\to c}(f(x))$ and $\lim_{x\to c}(g(x))$ do not exist, can $\lim_{x\to c}(f(x)g(x))$ exist?
  (c) If $\lim_{x\to c}(f(x))$ and $\lim_{x\to c}(f(x) + g(x))$ exist, must $\lim_{x\to c}(g(x))$ exist?
  (d) If $\lim_{x\to c}(f(x))$ exists and $\lim_{x\to c}(f(x) + g(x))$ does not exist, can $\lim_{x\to c}(g(x))$ exist?
  (e) If $\lim_{x\to c}(f(x))$ and $\lim_{x\to c}(f(x)g(x))$ exist, must $\lim_{x\to c}(g(x))$ exist?

Question 3 I can do this question ☐

- Let $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \sin(x) & : x \leq c \\ ax + b & : x > c \end{cases},$$

with $a, b, c \in \mathbb{R}$ and $b \geq c$. For which values of $a \in \mathbb{R}$, if they exist, is f continuous at $x = c$?

Question 4 I can do this question ☐

- Prove the following limits of functions at the given point directly from the ε-δ-definition of a limit.
  (a) $\lim_{x\to 0}\left(\frac{1-\sqrt{1-x^2}}{x^2}\right) = \frac{1}{2}$.
  (b) $\lim_{x\to 1}\left(\frac{x^3}{x-1} - \frac{1}{x-1}\right) = 3$.
  (c) $\lim_{x\to 3}\left(\frac{x^2+2x-15}{x-3}\right) = 8$.
  (d) $\lim_{x\to 2}\left(\frac{x^2-4}{x-2}\right) = 4$.

Challenge Qs

Now you are feeling more confident try these more challenging questions.

Question 5 I can do this question ☐

- For each of the following functions, prove directly from the ε-δ-definition of continuity that they are continuous on their entire domain
  (a) $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = |x|$.
  (b) $g : (0, \infty) \to \mathbb{R}$ given by $g(x) = \sqrt{x}$.
  (c) $h : \mathbb{R} \to \mathbb{R}$ given by $h(x) = e^x$.
  (d) $s : (0, \infty) \to \mathbb{R}$ given by $s(x) = \log(x)$.
  (e) $t : \mathbb{R} \to \mathbb{R}$ given by $t(x) = \cos(x)$.

Question 6 I can do this question ☐

- Prove Theorem 4.1 (iv) from Chapter 4 of the lecture notes.

Question 7 I can do this question ☐

- Let $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^{1/3}$.
  (a) Prove that f is continuous at $c = 0$.
  (b) Prove that f is continuous on $\mathbb{R}$.
  Hint: You may find the identity $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ *useful.*

Question 8 I can do this question ☐

- Prove that the function $g_a : \mathbb{R} \to \mathbb{R}$ given by $g(x) = x + a$ is continuous for all $a \in \mathbb{R}$ and each $x \in \mathbb{R}$.

Question 9 I can do this question ☐

- Prove Theorem 6.4 from Chapter 4 of the lecture notes.

Question 10 I can do this question ☐

- Prove Theorem 6.5 from Chapter 4 of the lecture notes.

Question 11 I can do this question ☐

- Prove Theorem 7.1 from Chapter 4 of the lecture notes.

Question 12 I can do this question ☐

- Finish the proof of the Extreme Value Theorem and Intermediate Value Theorem as given in the lecture notes.

Question 13 I can do this question ☐

- To gain a deeper understanding of the relationship between ε and δ in the definition of continuity, let's explore some minor variations of Definition 3.3 of the lecture notes. In each case, the function f is a real function with domain $\mathbb{R}$.
  (a) Let's say f is *onetinuous* at the point $c \in \mathbb{R}$ if for every $\varepsilon > 0$, we can choose $\delta = 1$ and it follows that $|f(x) - f(c)| < \varepsilon$ whenever $|x - c| < \delta$. Give an example of a function that is onetinuous on $\mathbb{R}$.
  (b) Let's say f is *equaltinuous* at point $c \in \mathbb{R}$ if for every $\varepsilon > 0$, we can choose $\delta = \varepsilon$ and it follows that $|f(x) - f(c)| < \varepsilon$ whenever $|x - c| < \delta$. Given an example of a function that is equaltinuous on $\mathbb{R}$ and nowhere onetinuous, or explain why there is no such function.
  (c) Let's say f is *lesstinuous* at point $c \in \mathbb{R}$ if for every $\varepsilon > 0$, we can choose $0 < \delta < \varepsilon$ and it follows that $|f(x) - f(c)| < \varepsilon$ whenever $|x - c| < \delta$. Given an example of a function that is lesstinuous on $\mathbb{R}$ and nowhere equaltinuous, or explain why there is no such function.
  (d) Is every lesstinuous function continuous? Is every continuous function lesstinuous? Explain your reasoning in both cases.

Question 14 I can do this question ☐

- Let $D \subset \mathbb{R}$. Then $c \in D$ is said to be an **isolated point** of D if there exists an ε-neighbourhood $V_\varepsilon(c)$ of c such that $V_\varepsilon(c) \cap D = \{c\}$.
  (a) Let $f : D \to \mathbb{R}$. Prove that if $c \in D$ is an isolated point, then f is continuous at c.
  (b) Hence prove that every sequence $a : \mathbb{N} \to \mathbb{R}$ is continuous on $\mathbb{N}$.

Question 15 I can do this question ☐

- Let $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = [[x]]$ with $[[x]] = \max(\{a \in \mathbb{Z} : a \leq x\})$. Prove that that f is discontinuous on $\mathbb{Z}$ but continuous on $\mathbb{R} \setminus \mathbb{Z}$.

Extension Qs

These questions are optional extension problems so you can gain more practice and confidence with these topics.

Question 16 I can do this question ☐

- Let $f : \mathbb{R} \to \mathbb{R}$ such that there exists a constant $c \in (0, 1)$ with

$$|f(x) - f(y)| \leq c|x - y|$$

for all $x, y \in \mathbb{R}$.

(a) Prove that f is continuous on $\mathbb{R}$.

(b) Pick some point $y_1 \in \mathbb{R}$ and construct a sequence by *iteration*, explicitly

$$(y_1, f(y_1), f(f(y_1)), \dots).$$

In general, if $y_{n+1} = f(y_n)$, show that the resulting sequence (y_n) is a Cauchy sequence. Hence, $\lim(y_n) = y$.

(c) Prove that y is a **fixed point** of f. That is, $f(y) = y$. Additionally, prove it is unique.

(d) Finally, prove that for any $x \in \mathbb{R}$, the sequence $(x, f(x), f(f(x)), \dots)$ converges to y, defined in part (b).

Question 17 I can do this question ☐

- Let $f : [0, 1] \to \mathbb{R}$ be continuous on $[0, 1]$ with $f(0) = f(1)$.

(a) Show that there exists $x, y \in [0, 1]$ satisfying $|x - y| = 1/2$ such that $f(x) = f(y)$.

(b) Show that there exists sequences (a_n) and (b_n) with $a_n, b_n \in [0, 1]$, for each $n \in \mathbb{N}$, such that $|a_n - b_n| = 1/n$ and $f(a_n) = f(b_n)$.

(c) Show that there does necessarily exist $x, y \in \mathbb{R}$ with $|x - y| = 2/5$ and $f(x) = f(y)$.

End Material

Review and Summary

Make sure you have fully understood and read the notes. Once the solutions for this worksheet have been released, check your answers and make a note of areas of concern. If you are still unsure then email your lecturer or drop by in office hours and we will be happy to help!

Completing Worksheets: Tips

- Proceed carefully and systematically from beginning to end making sure to note theorems and properties utilised.
- Work together with other students, discuss ideas and concepts.
- Prioritise carefulness over speed.
- Include enough information in your answers that you can easily understand your thinking when you come back to review the worksheets at a later date.
- Do ask for help if you get stuck!